Silicon Sampling for Indian Public Opinion

ABSTRACT / INTRODUCTION / MOTIVATION

Silicon sampling gives a language model a respondent's demographic profile and treats its answer as that person's survey answer. If it were faithful, surveys would become far cheaper.

We test this in India on **Pew Religion in India** (29,999 adults) and **WVS-7 India** (1,692 adults). Every method is scored on the same respondent-question rows as matched classifiers, and we check **direction**, **calibration** and **subgroup gaps**, not only accuracy.

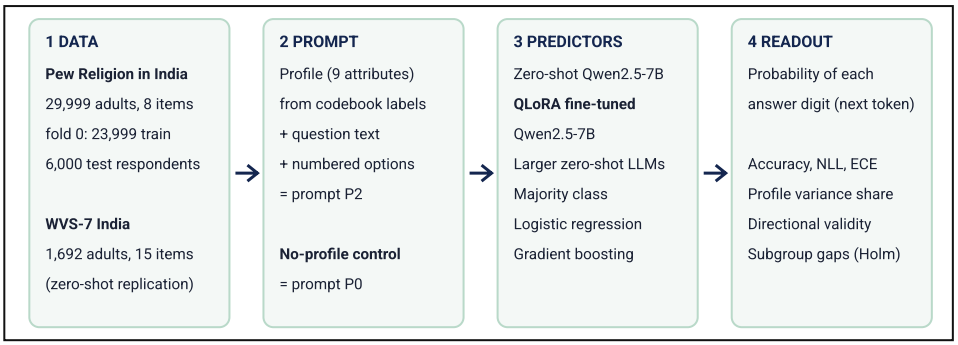


Image 01 : Study design: one prompt, identical rows for every predictor

LITERATURE REVIEW / BACKGROUND

- US attitudes are reproduced (Argyle 2023) but many groups misrepresented (Santurkar 2023).
- Individually, LLMs match or trail baselines and inflate group gaps (Chen 2026; Miranda 2025).
- Personas explain under 10% of variance (Hu 2024); fine-tuning can generalise (Suh 2025).

Gap: India, the profile's own contribution, transfer to unseen questions.

OBJECTIVES

RQ1 How does a prompted LLM use the profile? **RQ2** What does fine-tuning change versus classifiers? **RQ3** Does it transfer to unseen questions? **RQ4** Do size or more data matter?

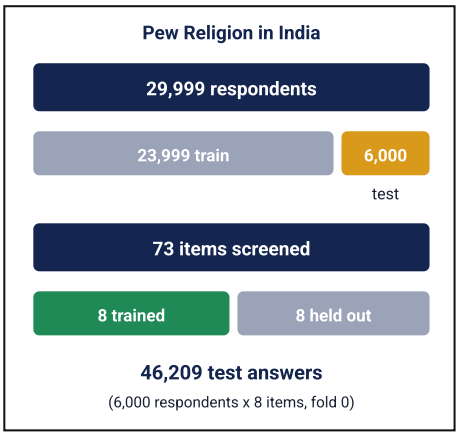


Image 02 : Pew Religion in India: respondents and items

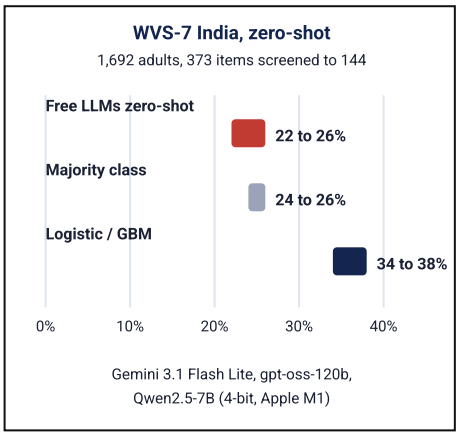


Image 03 : WVS-7 India: first zero-shot replication

METHODS

Data. Pew fold 0: 23,999 training and 6,000 test respondents, 8 items, 46,209 test answers; items 9 to 16 held out for transfer.

Prompt. Profile, question and numbered options in one fixed template; a no-profile control measures what the profile adds.

Predictors. Zero-shot Qwen2.5-7B, the same model fine-tuned with QLoRA, and majority, logistic and gradient-boosting classifiers on the same rows.

Readout. The probability of each answer digit at the next token.

```
You are a survey respondent with the
following demographic profile:
- Sex: Male
- Age: 18-25
- Marital status: Unmarried
- Education level: Senior secondary
- Income decile: Rs. 10,001 and above
- Religion: Hindu
- Urban/Rural: Rural
- Region: West
- Caste: General Category
Question: How many children have you had?
Answer options: 1. No Children
2. One Child 3. Two Children
4. Three or More Children
Respond with exactly one character:
1/2/3/4. No words, no explanation.
Your answer:
```

Image 04 : What the model sees: a real prompt

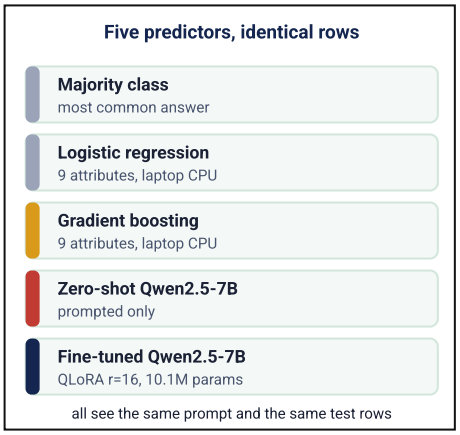


Image 05 : Five predictors on the same rows

METHODS (CONTINUED)

QLoRA fine-tuning. 4-bit NF4 base, LoRA rank 16 and alpha 32 on the attention projections, 10.1M trainable parameters, 184,853 examples, 5,777 steps, 6.6 hours on a university lab GPU.

Scoring. Kaggle 2 x T4 in fp16, about 0.2 s per answer and 75 minutes per condition. Classifiers train in under a minute on a laptop.

Reproducibility. Code, folds and prompts are in the project repository, and every figure on this poster is computed from stored model outputs.

DIAGNOSTICS

- Accuracy and negative log-likelihood
- Expected calibration error
- Profile share of prediction variance
- Directional validity: correlation and slope of group differences
- Jensen-Shannon divergence
- Subgroup gaps, Holm-corrected permutation tests
- Respondent-clustered bootstrap intervals
- Per-question and per-subgroup breakdowns

LESSONS FROM THE BUILD

- WVS region labels were wrong on every code until checked against the official annex; fixing it moved subgroup gaps by several points.
- The first prompt wording caused 13% refusals; the final wording gave 0%.
- An apparent collapse on one item was a decoding artefact, fixed by the next-digit readout.
- All reported numbers come from one training run; more seeds are future work.

GROUP 55 | BHANUJ BHALLA 70022300017 · ISHAAN BANSAL 70022300011
YASH BHENDE 70022300137 · KRISHANK AYYELLA 70022300136
MENTOR: MS. KHUSHBU CHAUHAN

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RESULTS & DISCUSSION

On Pew fold 0 (46,209 answers) the **fine-tuned model reaches 38.0%**, against 30.4% for always guessing the majority answer, 38.5% for logistic regression and **40.0% for gradient boosting**. Zero-shot Qwen2.5-7B scores **16.1%**.

Fine-tuning closes **79%** of the gap from the majority guess to gradient boosting, but stays 2.1 points below it.

Per question, the fine-tuned model beats the majority guess on all 8 items, by 1.3 to 24.8 points, and trails gradient boosting on every one.

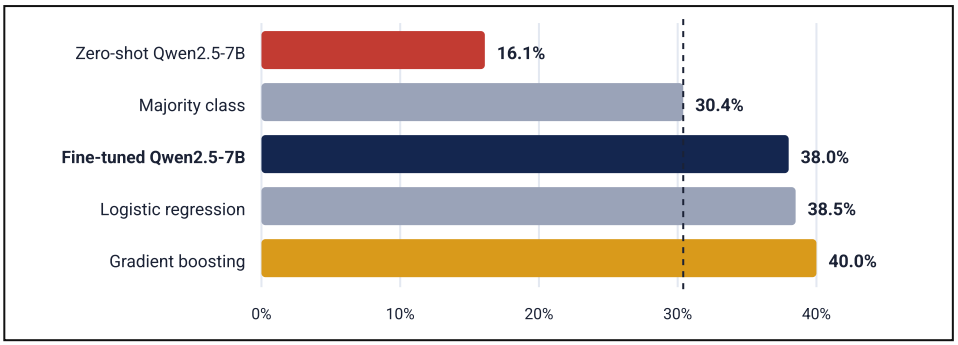


Image 06 : Accuracy on 46,209 test answers; dashed line is the majority guess

CALIBRATION AND CONFIDENCE

Calibration error is **0.009**: when the model says 60%, it is right about 60% of the time.

Above 70% confidence it is right **87.3%** of the time, on 1,338 answers, nearly all from the number-of-children question. The driver is marital status: unmarried respondents are 98.8% confident and 96.6% accurate.

A classifier finds the same rule, so this shows a learned pattern, not superiority. The model is strongest where the profile decides the answer and weakest on opinions.

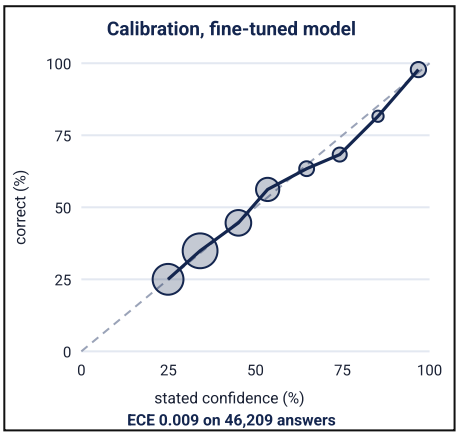


Image 07 : Stated confidence against accuracy

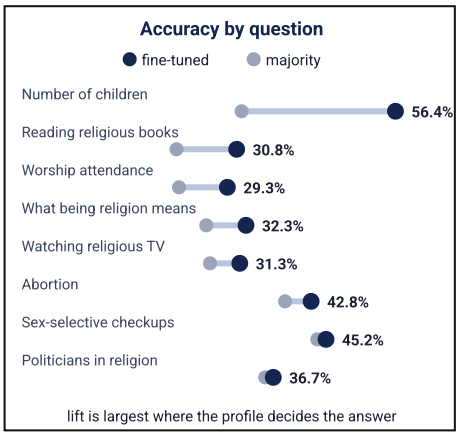


Image 08 : Accuracy by question against majority

PROFILE USE AND TRANSFER

Zero-shot, the profile explains 29% of the model's prediction variance against 4% for real answers, yet removing it raises accuracy by 3.6 points: the model reacts strongly and in the wrong direction.

Group differences correlate 0.22 (zero-shot), 0.58 (fine-tuned) and 0.64 (gradient boosting) with the truth.

On 8 unseen items the adapter scores 28.9% against 28.5% for the base model: **no detectable transfer**.

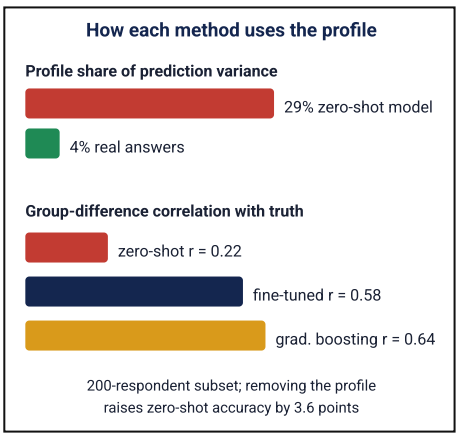


Image 09 : Profile share of variance, direction

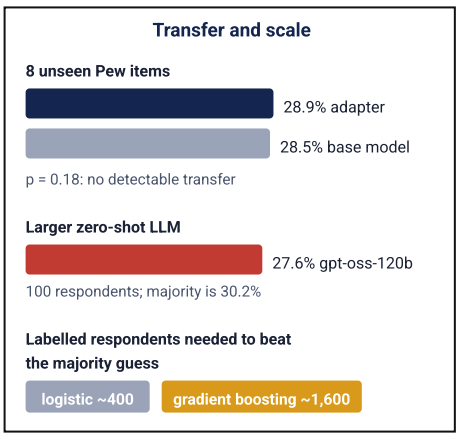


Image 10 : Transfer, a larger LLM, data needs

DISCUSSION

Prompting alone does not make a faithful Indian respondent. Fine-tuning gives a calibrated simulator close to, but not above, matched classifiers, at far higher cost.

Limits. One fold, one seed, one epoch; a small transfer test; survey weights not applied.

Next. All five folds and seeds, more held-out items, larger models on the same rows.

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project hyperlink : https://github.com/DevilHunter695/silicon-sampling-capstone

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